



Registers		
A	00	
BC	40 00	
DE	00 50	
HL	00 0C	
PSW	00 00	
PC	0015	
SP	FF FF	
Int-Reg	00	

Decimal - Hex Conversion

Decimal	Hex
0	0

I/O Ports

0	-	00
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Memory

0	-	00
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```
memory_alloc.asm
1 ; Memory Allocation Module
2 ; Divide Total Memory (16-bit) by Block Size (16-bit)
3 ; Compute Quotient and Remainder
4
5 LXI H, 4000H ; Total memory = 4000H (16384)
6 LXI D, 0020H ; Block size = 0020H (32)
7 MOV A, H
8 SUB H
9 SUB L, L
10
11 MVI B, 00 ; HL now = Total memory (DE preserved)
12 MVI C, 00 ; Quotient = 0
13 ; Remainder = 0
14
15 > DIV_LOOP:
16 MOV A, H
17 CMP D ; Compare high byte
18 JNCGE, DIV_DONE ; If Total < Block size, done
19 SUB D ; Subtract block size (high)
20 MOV H, A
21 MOV A, L
22 SUB E ; Subtract block size (low)
23 MOV L, A
24 INR B ; Quotient++
25 JMP DIV_LOOP
26
27 > DIV_DONE:
28 MOV A, B ; Quotient in B
29 STA 2100H ; Store quotient
30 MOV A, L ; Remainder in L
31 STA 2101H ; Store remainder
32 HLT
```

Start 2100h			
Address (Hex)	Address	Data	
2100	8448	00	; Quotient
2101	8449	00	; Remainder
2102	8450	00	
2103	8451	00	
2104	8452	00	
2105	8453	00	
2106	8454	00	
2107	8455	00	
2108	8456	00	
2109	8457	00	
210A	8458	00	
210B	8459	00	
210C	8460	00	

Line No Assembler Message

Memory allocation module executed.
Total Memory = 4000H (16384)
Block Size = 0020H (32)

Quotient = 512
Remainder = 0

Program assembled successfully